



TEMASEK
JUNIOR COLLEGE

TEMASEK JUNIOR COLLEGE
Preliminary Examinations
Higher 2

CANDIDATE
NAME

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CIVICS GROUP: _____/10

INDEX
NUMBER:

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BIOLOGY

9648/02

Paper 2 Core Paper

12th September 2011

2 hours

Additional Materials: Paper

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
TOTAL	

Section A

Answer **all** the questions in this section.

- 1** Plants native to warm climates sometimes suffer injuries when exposed to relatively low temperatures. For example, temperatures in the range of 10°C to 15°C can cause “chilling injuries” to some sub-tropical plants that normally grow between 20°C and 25°C. These chilling injuries may affect gene expression and reduce the rates of photosynthesis and protein synthesis. Some plants are more resistant to chilling injuries than others. These injuries could be due to damaged membranes.

Table 1.1 shows the fatty acid composition of mitochondrial membranes in various plants.

		Fatty acid composition / %					
		Chill-resistant plants			Chill-sensitive plants		
Fatty acid	No. of double bonds	Cauliflower	Turnip	Pea	Bean	Sweet potato	Maize
Palmitic	0	21.3	19.0	12.8	24.0	24.9	28.3
Stearic	0	1.9	1.1	2.9	2.2	2.6	1.6
Oleic	1	7.0	12.2	3.1	3.8	0.6	4.6
Linoleic	2	16.4	20.6	61.9	43.6	50.8	54.6
Linolenic	3	49.4	44.9	13.2	24.3	10.6	6.8
Ratio of unsaturated to saturated fatty acids		3.2:1	3.9:1	3.8:1	2.8:1	1.7:1	2.1:1

Source: A C Terry et al, *Plant physiology* 2000, 124, pages 183–190

Table 1.1

- (a) (i)** With reference to Table 1.1, describe the relationship between fatty acid composition and chill sensitivity.

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.....[2]

- (ii) Explain how the presence of double bonds affects membrane integrity at low temperatures.

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- (iii) Suggest how the function of mitochondria might be affected when a sweet potato plant is subjected to low temperatures.

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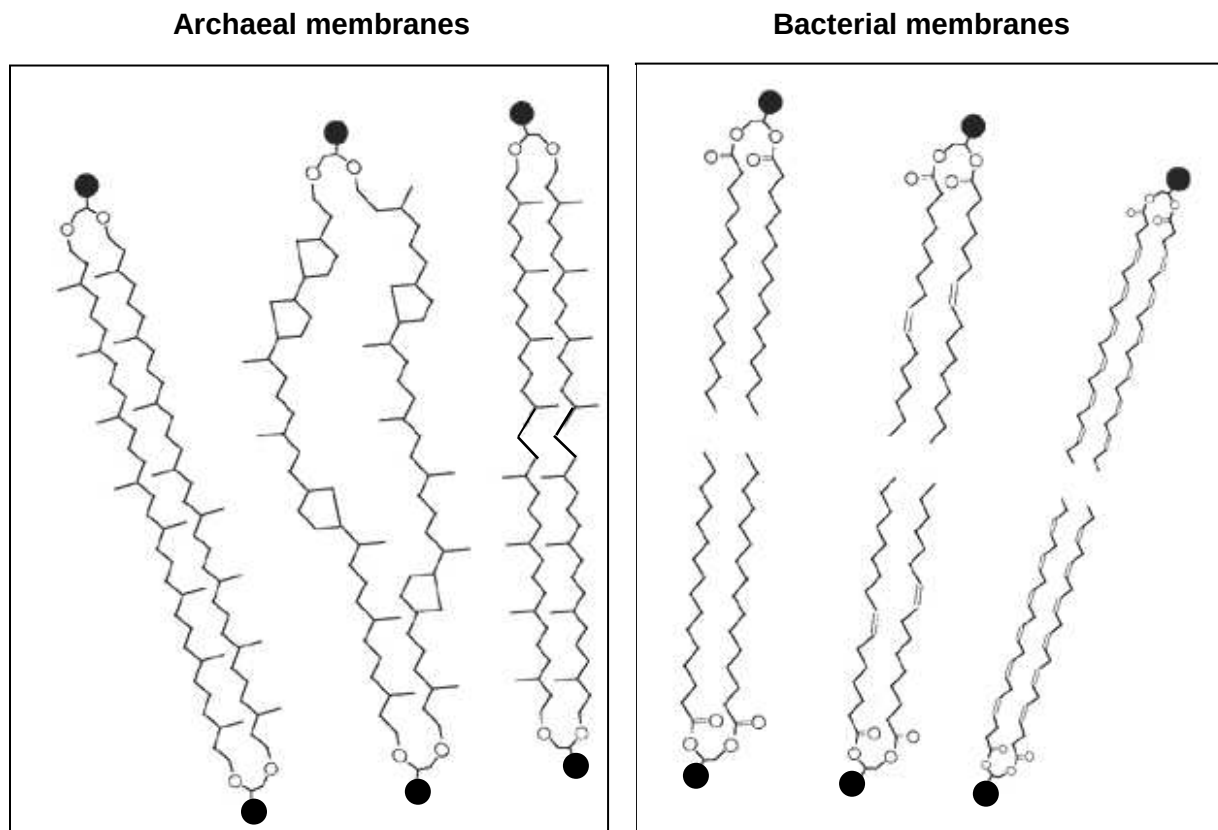
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.....[2]

In the late 1970s, a new group of organisms, the Archaea was discovered. These single-celled organisms were found to inhabit extreme environments, such as salt lakes, hot springs and volcanic vents at temperatures of well over 100°C.

Archaea were initially thought to be structurally similar to bacteria. However, further work by scientists revealed major differences in the chemical compositions of their cell membranes. For example, the hydrocarbon tails of archaeal phospholipids consist of isoprenoid chains which are long chains with multiple side branches and rings. This is in contrast to the fatty acid chains which make up bacterial phospholipids.

Fig. 1.1 shows the structure of the phospholipids in the membranes of archaea and bacteria.



Source: *Nature Reviews Microbiology* 5, 316-323 (April 2007)

Fig. 1.1

- (b) (i)** With reference to Fig. 1.1, other than the presence of side branches and rings, state two structural differences between the phospholipids of archaea and bacteria.

Archaea membranes	Bacterial membranes

[2]

- (ii) Suggest how the differences stated in (b)(i) enable archaea to thrive in environments with extreme conditions.

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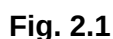
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[Total: 11]

- Fig. 2.1 shows the rate of photosynthesis (measured by rate of CO_2 fixation, graph drawn in solid line) and the rate of photorespiration (measured by rate of O_2 fixation, graph drawn in dotted line) in wheat over a typical summer day.



-[3]

(b) State the type of inhibitor that oxygen is to the enzyme RuBisCO.

.....[1]

Some plants have mechanisms to reduce photorespiration, such as the C4 plants. These plants undergo additional steps of capturing carbon dioxide using another enzyme (PEP carboxylase) to form oxaloacetate, before undergoing the Calvin cycle. PEP carboxylase has been found to have a lower K_m than RuBisCO for carbon dioxide. Examples of C4 plants include sugar cane and maize.

(c) Describe and explain the type of environment that C4 plants are usually found in.

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.....[3]

(d) Explain why Calvin cycle is still necessary in C4 plants.

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[Total: 9]

- 3 Back colour in the Kesametian Finch (Fig. 3.1) is determined by two alleles, blue and yellow. When pure breeding blue finches are mated with pure breeding yellow finches, the backs of the male progeny are always green, while the female progeny will have the same back colour as their fathers.



Fig. 3.1

- (a) Using suitable symbols, construct a genetic diagram to explain the results.

(b) Explain the above results.

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Female birds with yellow backs were mated with male birds with green backs. A number of geneticists then pooled their data on the results of the offspring.

Table 3.1 shows the number of offspring for each of the different phenotypes.

number of offspring	phenotype
28	male, green back
26	male, yellow back
28	female, blue back
18	female, yellow back

Table 3.1

(c) Apply the χ^2 test to determine whether the results obtained conformed to the expected Mendelian ratio for such a cross.

(i) Complete the table below, to show the observed results (O), the expected results (E) and the deviations of the observed from expected (O-E) for the 4 phenotype classes.

	Expected numbers	Observed numbers	O-E
male, green back			
male, yellow back			
female, blue back			
female, yellow back			

[1]

- (ii) Using the formula given below, calculate the value of χ^2 .

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

[1]

- (iii) The χ^2 value obtained indicates that the deviation from the expected results was statistically insignificant and due to chance. Explain how you would arrive at this conclusion with reference to the following χ^2 table.

Degrees of freedom	Probability, p (that the deviation is due to chance alone)				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47
5	9.24	11.07	13.33	15.09	20.51
6	10.64	12.53	15.03	16.81	22.46

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- (d) The cross was repeated with offspring from the F1 generation. Unexpectedly, one of the female offspring was found with a green back. Suggest a possible reason for the observed deviation.

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[Total: 12]

- 4 “Cancer is a disease of the genome, triggered by the accumulation of genetic errors that eventually transform a normal cell into a tumour cell. Such mutations might inactivate genes that normally oppose tumour development, or activate genes that drive cell growth or interfere with cell differentiation or death.

This leads to the 2 classes of genes which are responsible for cancer formation.”

Nature 417, 906-907 (27 June 2002)

- (a) (i) Explain what it means for cancer to be “triggered by the accumulation of genetic errors”.

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- (ii) With named examples, distinguish between the 2 classes of genes stated in line 5.

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Extracellular signals (growth factors) that activate one of two types of receptor — receptor tyrosine kinases and G-protein-coupled receptors — can result in the activation of RAS, leading to activation of BRAF and the downstream cascade.

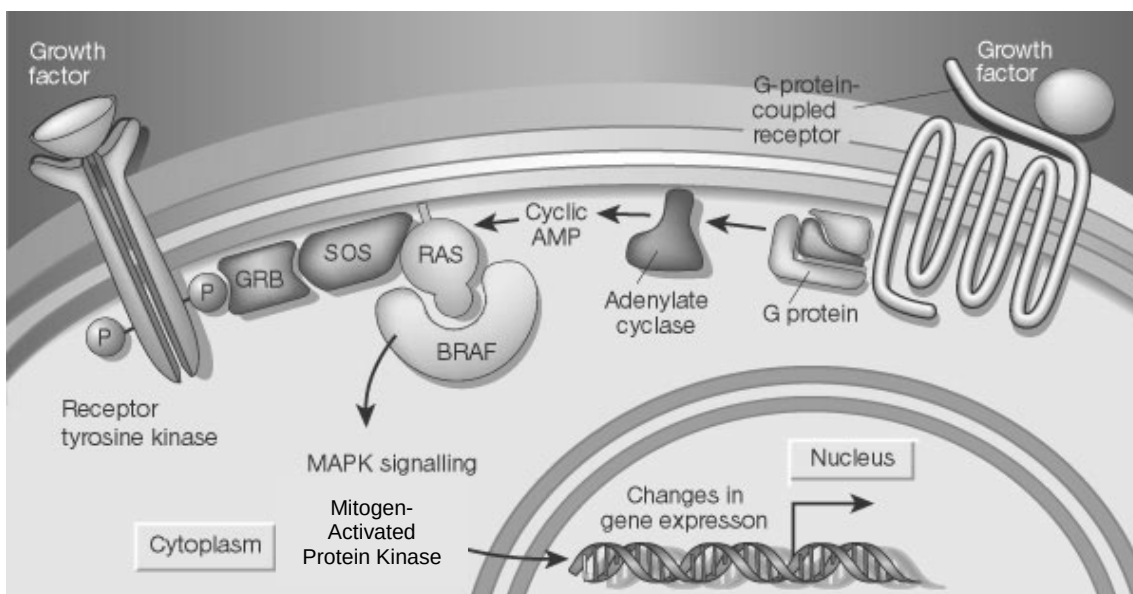


Fig. 4.1

(b) (i) With reference to Fig. 4.1, describe how *Ras* may contribute towards the development of cancer.

.....[3]

Besides *Ras* mutations, *Braf* mutations are also found in several cancer types. A base substitution in the *Braf* gene could result in a change in catalytic region of the protein, such that ongoing phosphorylation occurs even without activation of the BRAF. This causes the constitutive activation of the MAPK pathway.

- (ii) Suggest and explain how two other mutations of *Braf* could also potentially lead to the cancer.

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[Total:10]

- 5 University of Alberta researchers have unearthed a mysterious link between an ancient lizard found in Africa (33 million years ago) and the biggest modern-day lizard, the Komodo dragon (shown in Fig. 5.1), half a world away in Indonesia. This finding was initially challenged by other researchers on the basis of biogeography. However, there was other evidence discovered later that the ancient lizard had the ability to swim.



Fig. 5.1

- (a) Explain how the researchers could have deduced the relationship between the ancient lizard and the Komodo dragon.

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- (b) Explain why the finding was challenged on the basis of biogeography.

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[3]

- (c) Suggest how the Komodo dragon could have remained so similar to the ancient lizard.

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[1]

Another species of monitor lizard, known as Torch monitor, was recently discovered in Indonesia and found to be a close relative of the Komodo dragon due to many resemblances. One main difference between the species is their diet, where Torch monitor thrives on small animals, while the Komodo dragon preys on bigger invertebrates, birds and mammals.

(d) Describe how speciation could have occurred to give rise to the Torch monitor.

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(e) Different species concepts are used by biologists to define what a species is. Explain the consensus on the various species concepts.

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(f) Suggest one other way that biologists can determine the proximity of the two species.

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[Total:12]

6 Blood glucose level is regulated by the hormones glucagon and insulin.

(a) Explain how blood glucagon level is homeostatically regulated.

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Fig. 6.1 shows the interaction between insulin and its receptor.

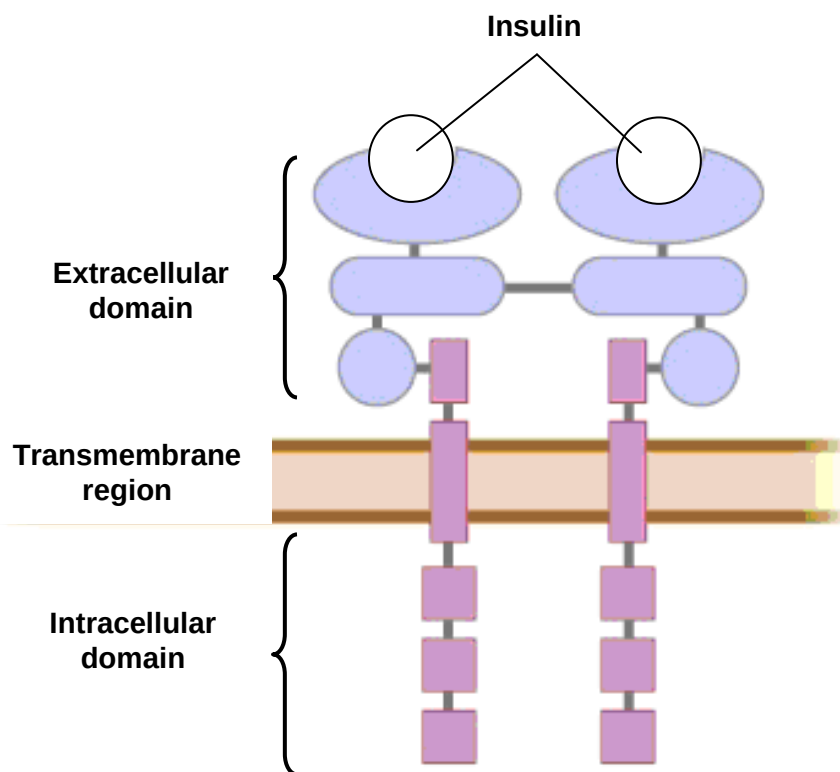


Fig. 6.1

(b) Relate the structure of the insulin receptor to its role in cell signalling.

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.....[4]

Fig. 6.2 shows the sequence of cell signalling events when insulin binds to its receptor.

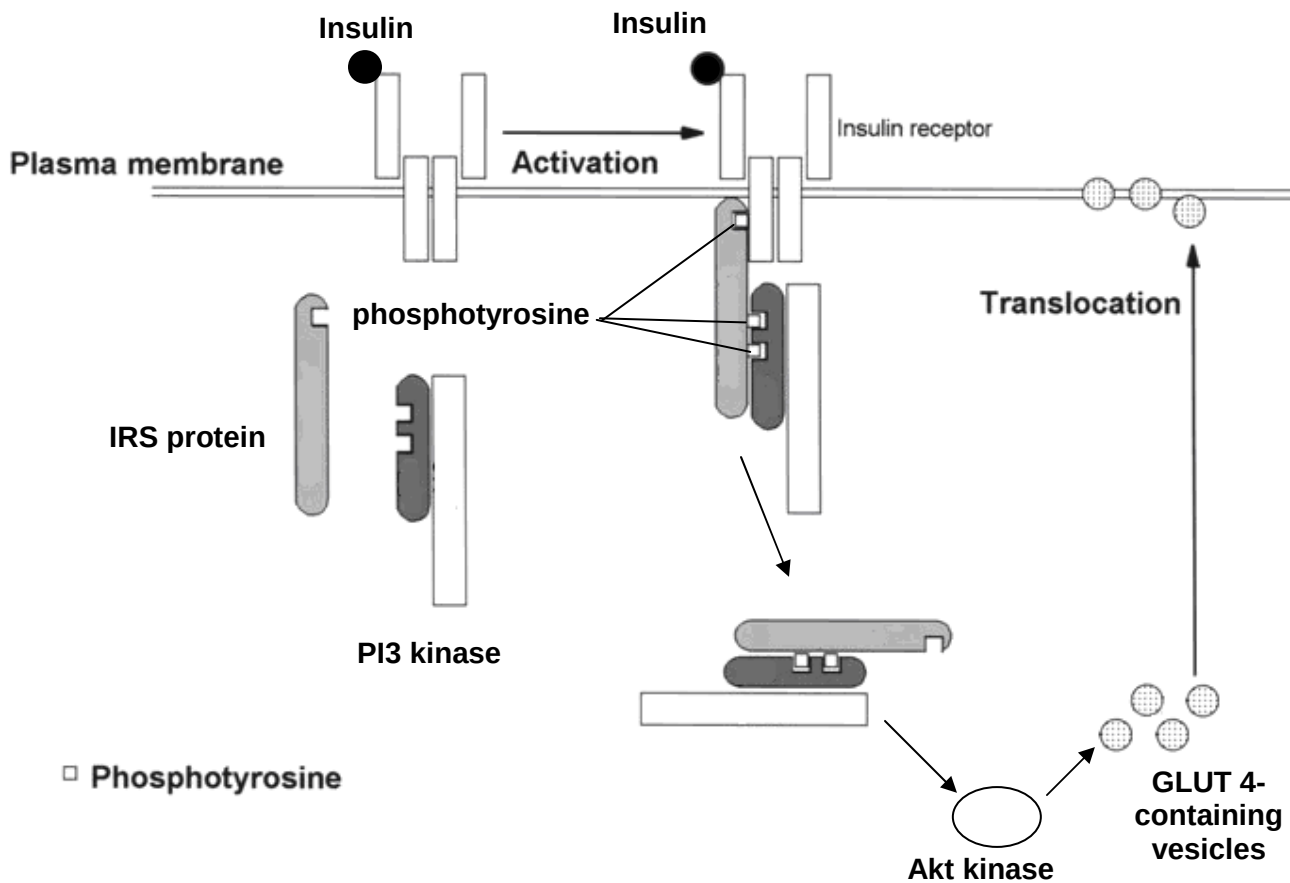


Fig. 6.2

[Turn over

- (b) (i)** With reference to Fig. 6.2, describe the signalling events that take place after the insulin receptor is fully activated.

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- (ii)** Explain the importance of signal termination and state one way in which this is achieved.

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[Total:13]

7 Fig. 7.1 shows the cross section of an axon.

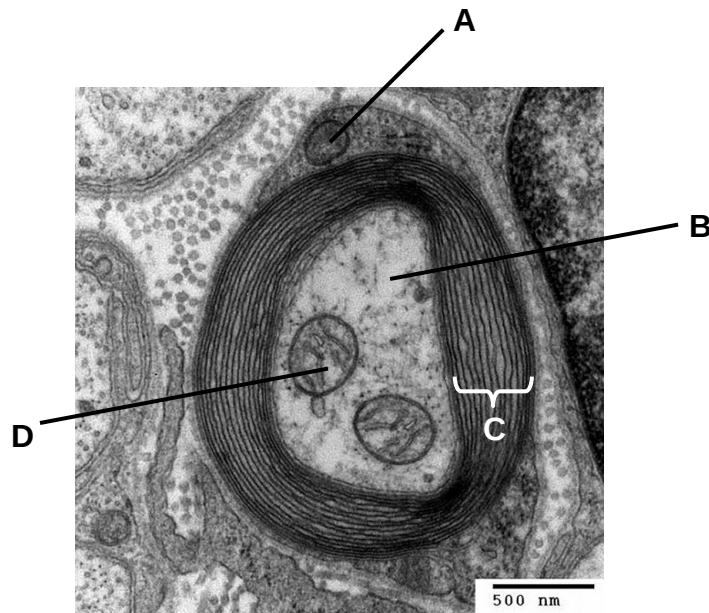


Fig. 7.1

(a) Identify the parts labeled A to D in Fig. 7.1.

A:

B:

C:

D:

[2]

The axon shown in Fig. 7.2 is typical of patients suffering from Multiple Sclerosis. The membrane potential along a myelinated axon is measured using microelectrodes at points P to U which corresponds to regions without a myelin cover. The bottom of the graph is a schematic diagram showing the relative positions of points P to U along the axon. (Note that microelectrodes at points T and U did not register a change in membrane potential and has been omitted from Fig. 7.2 for clarity purposes.)

[Turn over



Graph 1: Change in membrane potential at point P
Graph 2: Change in membrane potential at point Q
Graph 3: Change in membrane potential at point R
Graph 4: Change in membrane potential at point S

Fig. 7.2

(b) With reference to Fig. 7.2,

(i) describe and explain the change in membrane potential at point P of the axon.

[3]

(ii) describe and explain the change in membrane potential at point S.

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.....[2]

The neurones in the MS patients are different in that the antibodies for myelin are present in high concentrations, causing an autoimmune response that leads to partial degeneration of the myelin sheath.

(c) Explain why microelectrodes at points T and U did not register a change in membrane potential.

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.....[2]

Table 7.1 shows the diameter and transmission rate of action potential for different types of neurones.

Type of neurone	Presence of myelin	Average diameter (μm)	Average transmission rate (m/s)
Sensory	No	2.12	3.9
Sensory	Yes	4.05	8.4
Motor	No	1.89	2.9
Motor	Yes	4.20	10.9
Relay	No	2.03	3.6
Relay	Yes	3.89	8.1

Table 7.1

- (d) Using Table 7.1, explain how the presence of myelin and the diameter of the axon affects the transmission of a nerve impulse.

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.....[2]

In an anti-smoking campaign, the success of the campaign is measured by the number of people who pledged to quit smoking. As a matter of fact, to quit smoking requires more than willpower. Investigators attempt to understand and demystify the physiological effects of nicotine.

The first step was to trace the movement of nicotine by introducing radioactively labeled nicotine into the brain and see which type of carrier protein it binds to. To the surprise of many, it binds to the receptor directly (a subset of receptors for acetylcholine, on Fig. 7.3) on a post synaptic nerve cell.

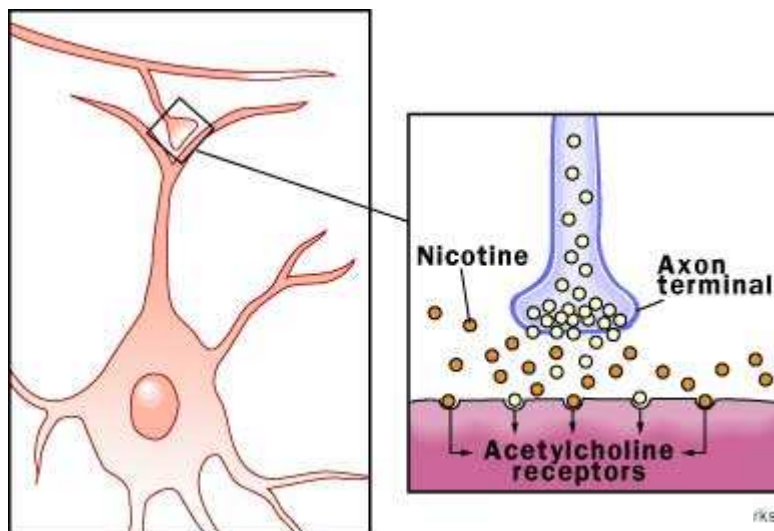


Fig. 7.3

- (e) (i) Suggest why nicotine could bind to the acetylcholine receptors.

.....[1]

- (ii) Suggest how nicotine affects the post synaptic membrane.

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Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a)** Discuss the roles of the different RNA molecules involved in translation. [6]
- (b)** With the use of named examples, compare between an inducible and a repressible operon. [8]
- (c)** Describe how meiosis brings about genetic variation for natural selection to occur. [6]
- 9 (a)** Describe the roles of centromeres and telomeres. [7]
- (b)** Describe how the Human Immunodeficiency Virus uses the host cell for its replication after entry and explain how it can then cause AIDS. [8]
- (c)** Outline the processes of binary fission and transduction in bacteria and briefly explain their significance. [5]